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RETROFIT PLAN · IMPROVEMENT OPTIONS EVALUATION & MEDIUM-TERM IMPROVEMENT PLAN

Anonymised sample report

A whole-house plan for a warmer, drier and cheaper-to-run home, built from the thermal imaging, drone roof survey, ventilation tests and room-by-room inspection carried out on 20 October 2026.



PREPARED FOR	VISIT	ISSUED	ASSESSOR
The client	Tuesday 20 October 2026, 9.30am	20 October 2026	Dominic Bowkett

01 At a glance

Thank you for having me on Tuesday. As you said, you know where the house is cold; this plan sets out why, and what to do about it. It brings together the thermal images, the drone survey, the ventilation tests and the room-by-room inspection, and turns them into a sequence of improvements, starting with quick wins for this winter and leaving the bigger jobs until there is a natural moment to do them.

Your house is a solid-brick Edwardian mid-terrace in fair condition. The energy assessment on the day rates it **69 (C)**, with a potential of 85 (B). The heating works well and the roof is generally in good order. The cold you have felt downstairs comes mostly from **air leaking in** and from a few **uninsulated weak spots** (the bay, the floor and the porch), rather than from a shortage of heat. As I said on the day, I have written the plan from a "top-spec" energy point of view: it shows everything that would make a difference, in a sensible order, but you choose what to do and when. Every photo and thermal image from the visit is in the appendix at the end.

What I found

01 **The draught under the front-room window is real. It comes from the floor and the bay.**
The front of the ground floor is suspended timber over a ventilated void, with an air brick under the bay. There is an open gap under the skirting boards in the through-lounge and porch, and the thermal camera shows cold air lying along the skirting. The bay itself is almost certainly uninsulated: the wall beneath the bay window is the coldest surface in the house, at 13.4–14.2°C against 16–17.5°C on the wall above it.

02 **The windows and doors are older than they look.**
They are pre-2002 double glazing, without the low-emissivity coating or "warm edge" spacer bar that modern units have, and the uPVC frames have aged and are now likely to have air leakage. Replacing them is the biggest single improvement to comfort in the longer term, although it is expensive.

03 **The porch is a cold spot.**
One porch wall is a single thin skin of brick, and the porch roof showed cold on the thermal images, so there is probably little insulation above it. When the inner door is open, warm air from the house is drawn out through the porch roof and walls.

04 **The house has almost no working extract ventilation.**
The bathroom fan comes on with steam, but I measured only 0.39 litres per second against at least 15 l/s needed. The opening rooflight helps when it is used, but it cannot be left open in all weathers. The kitchen only has a recirculating cooker hood, which filters grease but does not remove moisture, and the internal doors are too tight to the floor for air to move between rooms. This needs fixing before any draught-proofing or insulation, so that the house stays dry as it gets tighter.

05 **The loft has very little insulation.**
It is boarded and used as an office and store, with the boiler and a radiator up there. The only insulation visible is a thin foil layer, around a millimetre thick, on the rafters. Whatever is under the boards could not be seen and is assumed to be around 100 mm, well below today's standard of 270–300 mm.

06 **Your heating is in good order, and the radiators are not the main problem.**
The downstairs radiators look small, but they are double convectors, so each gives out roughly as much heat as a single radiator twice the size. They are reasonably sized for the rooms. The cold downstairs is about heat escaping, not a lack of it.

07 **The roof is generally in good condition.**

The drone survey showed a small section of flashing coming away at the front, on one side and possibly the other, plus routine maintenance: lichen and moss, a few chipped tiles, and worn mortar on the party parapet and around the chimney. There were old moisture marks in the loft, but it felt dry; they may be linked to the flashing.

The five things I would do first

#	ACTION	INDICATIVE COST
1	**Fit proper extract ventilation**: replace the bathroom fan and add one to the kitchen, which only has a recirculating hood, using quiet, continuously running low-energy fans (dMEV) ducted to outside. Trim the internal doors to leave a 10 mm gap.	£450–£1,000
2	**Insulate under the bay window**: rigid foam board (such as Celotex) cut to fit on timber battens, with a small gap to the wall, plasterboard over and every joint sealed. A quick win for the coldest part of the room.	£150–£400 DIY £400–£1,000 fitted
3	**Insulate the porch roof**: lift the hatch and lay around 300 mm of loft insulation. It is a small space and a straightforward DIY job.	Low cost, DIY
4	**Seal the draughts**: the gap between floor and skirting in the lounge, hall and porch, the window frames to the plaster, and a threshold seal and letterbox brush on the front door.	£90–£270 DIY £500–£1,000 fitted
5	**Re-fix the loose flashing** at the front of the roof, and have the roofer check the parapet capping and chipped tiles while they are up there.	£250–£800 including access

Costs throughout this plan are indicative 2026 ranges for the South East, to help with budgeting. They are not quotes.

02 Your questions, answered

When you first got in touch you asked about insulation, windows and doors, draughts, radiator sizing and thermal bridging, and in your email before the visit you mentioned the draught under the front-room window and wondered about a bigger radiator. Here are straight answers to each.

“There is a draught under the front-room window. Where is it coming from?”

Mostly from the floor. The front of the house has a suspended timber floor with a ventilated void underneath, which is how it should be built: the void needs air moving through it to keep the joists dry. The problem is that some of that air is getting into the room. There is a visible gap between the new oak flooring and the skirting board, with dark dust lines along it, which is the tell-tale sign of air moving through. The thermal images show the coldest band of the room running right along the skirting under the bay.

The second part is the bay itself. The low wall under the bay windows (the “stall riser”) is a thin section of wall, and it was 13.4–14.2°C on the inside face, against 16–17.5°C for the main wall above. It loses heat faster and feels cold to sit near, which adds to the sense of a draught. There is also an unsealed joint between the window frame and the plaster in the bedroom bay above, and the same detail is worth checking downstairs.

There is also an air brick under the bay. It ventilates the floor void and must stay open, but with no insulation behind the bay wall it adds to the cold in that corner.

What to do: seal the skirting-to-floor gap with a flexible sealant or a skirting draught strip, seal around the bay window frames, and insulate under the bay window with rigid foam board on battens behind plasterboard. That last one is the quick, high-impact win for this room (Stage 1 in section 11).

“Should I get a bigger radiator in the front room?”

It is not the first thing I would spend money on. The radiators look small, but they are double convectors: two panels with fins between them, so each gives out roughly as much heat as a single-panel radiator twice its size. They are reasonably sized for the rooms. A bigger radiator would put more heat in, but with the draughts and the uninsulated bay it would be heating air that leaks straight back out.

Deal with the draughts, the bay and the porch first. If the room still struggles after that, modern radiators are more efficient, and you have the wall space to go taller or wider without moving much pipework, which your stepdad could take on.

“What about insulation?”

In order of value for your house: **under the bay window** and the **porch roof** (both small, cheap and quick), then the **loft** top-up, the **porch wall**, and the **suspended ground floor** in the through-lounge. The floor is the most disruptive, because the boards have to come up or the work is done from below, so most people leave it until they are replacing the floor anyway. For the **sloping roof over the bathroom**, it is impossible to know whether it has been insulated in the past without opening it up, so check and insulate it when it is next re-roofed. Insulating the main solid walls is possible, but it is expensive and disruptive and belongs at the end of the plan.

“Do the windows and doors need replacing?”

In the medium to long term, yes. They are pre-2002 double glazing. Newer units have a dark “warm edge” spacer bar between the panes and a low-emissivity coating on the glass that reflects heat back into the room; yours have neither. The uPVC frames have also aged and are now likely to have air leakage around the edges, which the thermal images point to, especially at the front door and the patio doors. Modern windows and doors would stop a lot of the draughts and keep more heat in.

It is a big cost, so it is a longer-term project. Until then, sealing the frames to the plaster, a threshold seal on the front door, and cleaning and re-brushing the patio-door seals will help. When you get quotes, use installers registered with TrustMark and FENSA (or CERTASS), get more than one quote, and be wary of “today only” discounts.

“What is thermal bridging, and do I have it?”

A thermal bridge is a spot where heat finds an easier way out than through the rest of the wall: a lintel, a thin section of wall, or a junction between materials. In your house the clearest examples are the bay stall riser, the lintel over the bedroom bay window, and the ceiling-to-wall junctions upstairs where the loft insulation stops short of the eaves. These are the places where condensation and mould would form first if the house were under-ventilated, which is another reason the extractor fans come first. Whenever insulation is added later, the detailing at these junctions matters as much as the insulation itself.

“Was the thermal imaging worth doing?”

Yes. The conditions were good enough to give clear, usable readings indoors, and it confirmed the source of the draught you had noticed. Section 04 shows the images.

03 The house

Type	Two-storey mid-terrace house, built around 1900–1929, with a two-storey canted bay at the front, a brick porch with a tiled lean-to roof (single thin brick wall, little or no roof insulation), a two-storey rear outrigger (kitchen with the bathroom over) and a single-storey rear extension with sliding patio doors.
Walls	Main house solid brick, 260 mm measured, painted textured render to the front and smooth painted render to the rear. Outrigger solid brick, around 300–320 mm, dry-lined. Single-storey rear (1967–75) cavity wall, around 300–320 mm, dry-lined, no cavity fill recorded. Porch (1976–82) single-skin brick. No wall insulation.
Roofs	Main roof pitched, concrete interlocking tiles, shared parapet to No 94 and a shared chimney stack. Outrigger pitched with artificial slates, a rooflight over the bathroom and a sloping bathroom ceiling. Single-storey rear: low-pitched, artificial slates behind a boundary parapet.
Loft	Fold-down ladder. Boarded throughout and used as an office and store. Foil multifoil insulation stapled to the underside of the rafters. Gas boiler and a radiator in the loft. Depth of insulation under the boards could not be seen; the 2025 EPC records 100 mm.
Floors	Suspended timber in the main house (engineered oak finish), solid in the outrigger and rear extension. No floor insulation.
Windows and doors	uPVC double glazing throughout, pre-2002 (no low-E coating or warm-edge spacer), including the rear sliding patio doors. Double-glazed rooflight over the bath. uPVC front door with glazed panel. No FENSA certificate is registered for the property.
Heating and hot water	Worcester Bosch Greenstar 25i ErP condensing combination boiler (2015 or later) in the loft, with a vertical flue through the roof; radiators with thermostatic valves; programmer and Worcester Comfort II RF wireless room thermostat in the through-lounge. Hot water on demand from the combi; one bath and one mixer shower. Smart gas and electricity meters fitted. All 23 fixed light fittings are LED.
Ventilation	Intermittent ceiling extractor in the bathroom (tested, not working effectively). Kitchen has a recirculating cooker hood only (no extract to outside). Trickle vents only in the rear living-room window (about 150 mm) and bedroom 2 (about 300 mm). Internal door undercuts below the 10 mm needed.
Floor area	64 m² internal, as measured on the day. Four habitable rooms, all heated.

Designations	Not listed and not in a conservation area. Inside a Smoke Control Area and the local council's HMO Article 4 area (neither affects this plan). No planning history on the council register.
Orientation	Front faces south-south-east onto the road, so the front roof slope gets good sun.

How you use the house

Two adults live here, with no vulnerabilities or disabilities. You moved in during the summer of 2025, and through the first winter you could not get downstairs to a comfortable temperature, while upstairs was fine. In winter you keep the inner porch door closed, although there are still gaps around it. The loft is used as a working office. There is a detached garage at the end of the garden, which is not part of this assessment; you mentioned its roof needs replacing.

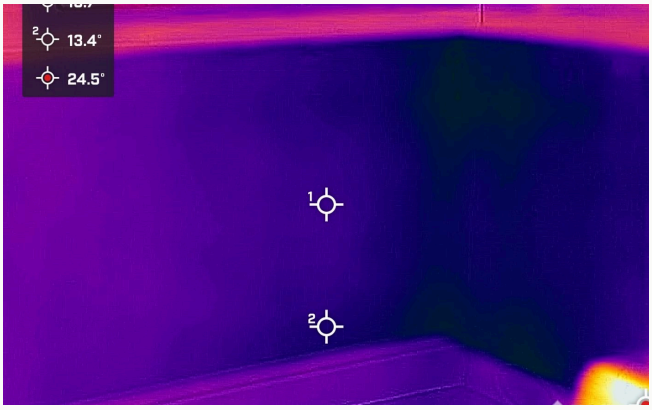
You are not sure how long you will stay, so the plan separates quick, cheap wins that pay back within a winter or two from longer-term upgrades that make most sense if you are staying, or that add value when you sell.

04 What the thermal images show

The images were taken between 9.32am and 9.42am with a FLIR Edge Pro, after the heating had been on since early morning at your normal setting. It was a fully overcast morning, about 12.5°C outside, with a light breeze (around 15 km/h) and a trace of drizzle in the area (modelled weather data for the address). With the house at around 20–21°C inside, the temperature difference was about 8°C, a little under the 10°C that is ideal, so the indoor images are reliable for spotting cold areas and draughts, and the outdoor ones are indicative. The colours show surface temperature: yellow and orange are warm, purple and black are cold. The numbers in the corner of each image are the readings at the marked spots. Thermal images show where heat is being lost and where surfaces run cold; they point to causes rather than proving them, so I have read them alongside what I saw in each room.



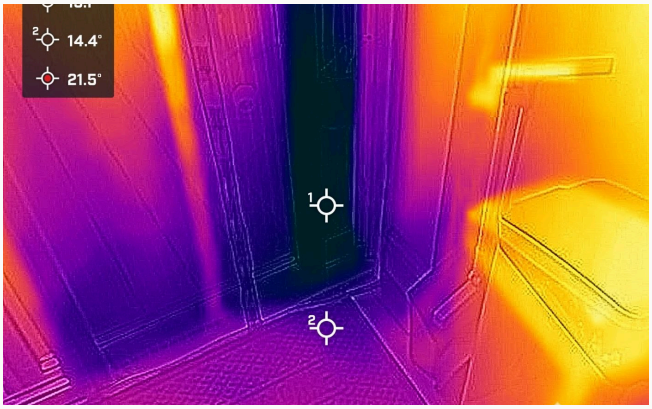
Through-lounge bay, from inside. The low wall under the bay windows is the darkest area, and a cold band runs along the skirting. The bright spots on the floor are plug-in appliances.
FLIR 09:32:31 · 16.0 / 17.5°C at the markers



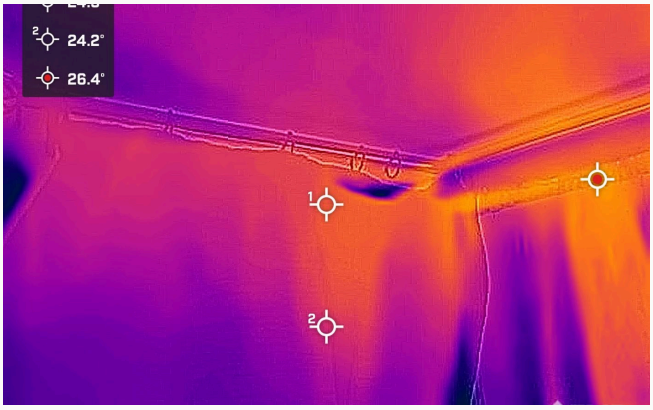
Under the bay window, close up. The stall riser was 13.4–13.7°C, the coldest wall surface in the house, against 16–17.5°C for the main walls. This is where the draught is felt.
FLIR 09:32:45 · 13.7 / 13.4°C



Front door threshold. A clear cold line along the bottom of the door, where outside air is getting in under it.
FLIR 09:33:45 · 16.3 / 16.4°C



Sliding patio doors. Cold air along the frame edges and the bottom track. Glass reflects infrared, so the 13–14.5°C readings on the glass are indicative only; the cold edges are the reliable part.
FLIR 09:34:41 · 13.1 / 14.4°C



Bedroom 1 bay, window head. A cold spot at the centre of the window head, where the lintel bridges the wall. Combined with the open joint at the reveal (section 07), this bay leaks heat at the edges.
FLIR 09:36:13 · 24.9 / 24.2°C (radiator below)



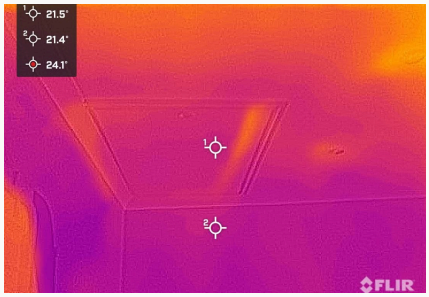
First-floor ceiling junction. A cooler band where the ceiling meets the outside wall, typical of loft insulation that stops short of the eaves.
FLIR 09:37:00 · 20.6 / 21.9°C



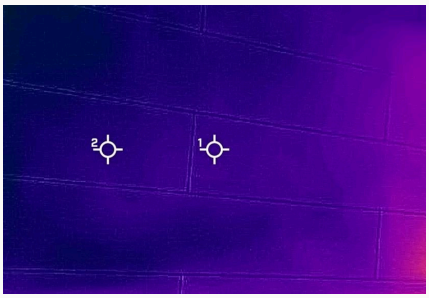
Lounge bay, wider view. The cold band runs along the whole base of the bay.
FLIR 09:33:07 · 19.1 / 18.1°C



Inside the front door. Cold line along the threshold and the bottom of the door.
FLIR 09:33:32 · 15.6 / 15.8°C



Ceiling with loft hatch. A cooler strip along the hatch edge, where it is not sealed or insulated.
FLIR 09:36:32 · 21.5 / 21.4°C



Bathroom wall. The tiled wall at 18.7–18.9°C; the warm line at the bottom is heating pipework.
FLIR 09:35:18 · 18.9 / 18.7°C



Bathroom floor. 16.3–16.4°C, with the radiator off at the time of the visit.
FLIR 09:37:16 · 16.4 / 16.3°C



Side return, outside. Warmth showing round the edges of the outrigger window frame.
FLIR 09:42:20 · 16.1 / 15.5°C, hot spot 16.9°C

On the positive side, the main ceilings upstairs read evenly at around 21.5°C, which suggests the loft insulation is reasonably continuous across the middle of the ceilings; the losses are at the edges. The loft hatch showed as a cooler strip.

05 Roof and exterior

I flew the drone over the main roof, the outrigger and the rear extension, and inspected the walls from the ground. Each item is marked ACT NOW (before winter or before other work), PLAN (within the next year or two, or at the next opportunity) or MONITOR.

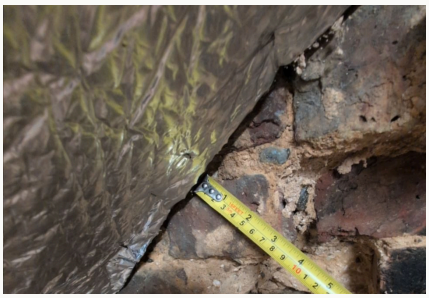


Front abutment flashing. A section of the lead flashing has come away at the side abutment, with a cracked tile beside it; the other side should be checked too. Re-fix and seal. ACT NOW



Side return. A sub-floor air brick sits at deck level and is partly blocked by the decking. The floor void needs this airflow. ACT NOW

Inside the loft



Party wall. Loose, eroded mortar and dark staining on the exposed brickwork below the roof line. MONITOR



Multifoil under the rafters. A thin foil layer, around a millimetre thick.

The staining on the party wall could be old soot from the shared flue or water tracking in from the parapet. It is worth having a look after a spell of heavy rain. If it is damp, the flashing and parapet capping repairs in Stage 0 are the likely fix.

06 Ventilation and moisture

A house needs a steady supply of fresh air and a way to get rid of the moisture from cooking, washing and breathing. In an older, leakier house a lot of this happens by accident through the gaps. Once you start draught-proofing, the deliberate ventilation has to be working properly, or moisture builds up on the coldest surfaces as condensation and mould. That is why ventilation sits at the front of this plan.

ITEM	FOUND	WHAT IT SHOULD BE	RESULT
Bathroom extract Ceiling fan, humidity-triggered	Comes on with steam, and sometimes on its own. Measured running after the shower: 0.14 m/s, **0.39 l/s** (Testo 417 with flow hood)	At least 15 l/s (intermittent) or 8 l/s (continuous), Approved Document F	FAIL

Kitchen extract	Recirculating cooker hood only; nothing extracts to outside	At least 30 l/s from a ducted cooker hood, or 60 l/s from a wall fan, or 13 l/s continuous	FAIL
Bathroom window	Opening rooflight over the bath	An opening window for purge ventilation (airing after a bath or shower)	PASS
Door undercuts	Measured at the internal doors: none gives a 10 mm gap. The kitchen door passes measured from the kitchen side but not from the through-lounge, where the oak floor sits higher.	About 10 mm above the floor finish so air can reach the extract	FAIL
Background ventilation	Trickle vents in only two windows: the rear living-room window (about 150 mm) and bedroom 2 (about 300 mm). None in bedroom 1, the front bay or the kitchen.	Trickle vents or equivalent in each habitable room, kept open	PARTIAL
Sub-floor ventilation	Air bricks under the bay and at the back of the house, ventilating the floor void. The rear one sits at deck level; the planters sit against the bay.	Clear cross-ventilation under the suspended floor	CLEAR



Bathroom fan test. 0.39 l/s, effectively no extraction.

What I recommend

- **Bathroom:** replace the fan with a continuously running, low-energy decentralised extract fan (dMEV) that ticks over quietly and boosts automatically when humidity rises. It is meant to run all the time, so you will not need to switch it off; it should have an isolator switch outside the bathroom for servicing, and the grille just needs a wipe every few months. Check the duct run to outside is short, insulated where it passes through the loft or cold void, and ends in a proper wall or roof terminal. Running cost is typically a few pounds a year.
- **Kitchen:** the recirculating cooker hood filters grease and smells but puts the moisture straight back into the room. Fit a dMEV on an outside wall of the kitchen, or a cooker hood ducted to outside (not a recirculating one), and use it whenever cooking. The grille on the outrigger wall should be given a proper cowl so moist air is thrown clear of the render.
- **Doors:** trim the internal doors to leave about 10 mm above the floor finish, measured on the side with the higher floor. Without that gap the fans cannot draw air through the house. The kitchen door needs it measured from the through-lounge side, where the oak floor is higher.
- **Trickle vents:** keep the two you have (rear living-room window and bedroom 2) open, especially overnight. Bedroom 1 has none, and it is where two people sleep with the door closed, so add one there; a vent in the front bay window would also help. Most uPVC frames can have them retro-fitted, and new windows (Stage 2) should come with them.
- **Floor void:** keep the air bricks under the bay and at the back of the house clear of decking, soil and planters. You will feel a little draught in the deep cupboard at the back; that is the void being ventilated, and it is expected. Never block these when draught-proofing: you seal the gaps *inside* the room, not the vents *outside*.
- **Kitchen reveal:** once the extract is working, strip the loose paint, let the plaster dry, and redecorate with a breathable paint.

The air purifier and dehumidifier in the lounge are doing useful work in the meantime. With working extract fans and fewer draughts you should find you need the dehumidifier much less.

07 Draughts and air leakage

These are the gaps I found that let cold air in. Most can be sealed in an afternoon with a tube of flexible sealant, some draught strip and a brush seal, and together they are likely to be the single biggest improvement to comfort downstairs.



Through-lounge skirting. An open gap between the oak flooring and the skirting, with dark dust lines where air has been moving through from the floor void.



Porch and hall. The same gap at the skirting, here with the edge beading lifting.



Bedroom 1 bay. An open joint running the full height of the reveal where the window frame meets the plaster. Check the lounge bay for the same.

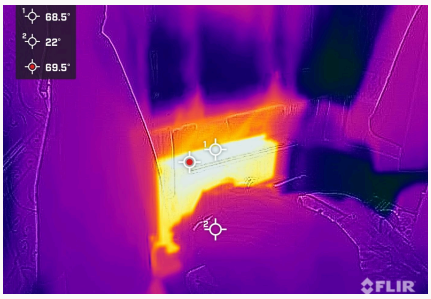
WHERE	FIX	INDICATIVE COST
Skirting to floor, lounge, hall, porch	Flexible acrylic or silicone sealant, or a skirting draught strip. Seal holes where pipes and cables go into the floor (including the gas meter cupboard).	£50–£150 DIY
Bay windows, up and down	Seal the frame-to-plaster joint inside with a flexible decorator's sealant; check and renew the mastic outside.	£30–£80 DIY
Front door	Threshold brush or compression seal, letterbox brush flap, check the keeps so the door pulls tight.	£25–£60 DIY
Loft hatch	Compression seal round the frame, insulation board fixed to the top of the hatch.	£30–£80 DIY
Patio doors	Clean the tracks, replace brush pile seals, adjust rollers.	£40–£150

> **Do not seal:** air bricks, the extract fan outlets, the gaps under internal doors, or any vents serving the boiler. Draught-proofing is about stopping *uncontrolled* air; the controlled ventilation must stay open.

08 Heating and hot water



Worcester Bosch Greenstar combi in the loft, with a vertical flue through the main roof.



Bedroom radiator at 66–70°C, heating evenly across the panel.



Gas meter in the lounge cupboard.

The system is in good working order. The radiators I imaged were heating evenly with no cold patches, and the controls (programmer, wireless room thermostat and radiator valves) are the right ones. A few low-cost changes will make it work better.

- **Turn the boiler's radiator flow temperature down** to around 55–60°C. The radiators were running at 65–70°C on the day. A condensing boiler is most efficient when the water coming back to it is below about 55°C, and turning the flow down often saves 6–8% on gas with little change in comfort. If the house takes too long to warm up, nudge it back up a little at a time.
- **Have the system balanced** at the next service so the downstairs radiators get their fair share of flow before the upstairs ones.
- **Lag the pipework and condensate pipe in the loft.** The loft is outside the insulated part of the house, and a frozen condensate pipe is the most common winter breakdown for combis in lofts.
- **Keep the annual service** going. The Greenstar 25i is a modern condensing boiler (2015 or later) with plenty of life left, so there is no need to replace it early; plan the heat pump for when it does need replacing.
- **The downstairs radiators** are double convectors and reasonably sized for the rooms. Upgrading them is optional and comes after the draughts and insulation; if you do, go taller or wider on the same pipework, sized from a room heat loss calculation (see section 02).
- **The bathroom radiator** was off during the visit, which, with a sloping ceiling that may have little or no insulation, makes the bathroom the coldest room upstairs and more prone to condensation. Keep it on low in cold weather.
- **The loft radiator**: if the loft is to stay as storage, turn it off and insulate the floor (Stage 1). If it is to be a room, it needs to be insulated as one first (Stage 3).

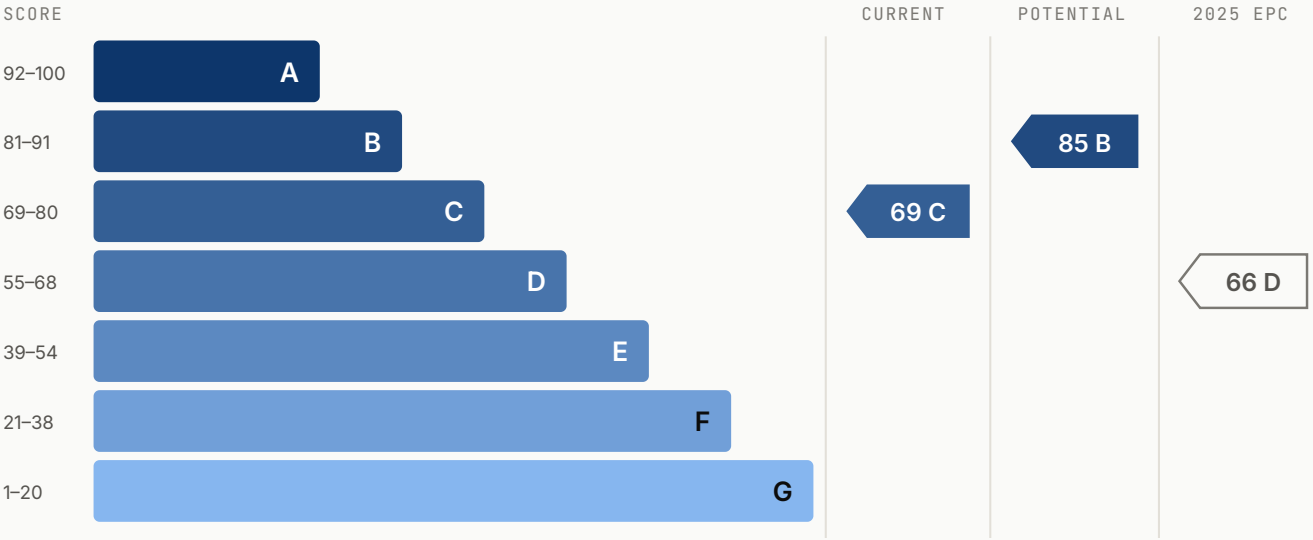
09 Energy performance: the baseline

Alongside the inspection, I carried out a full energy assessment of the house on the day, using the same government method (RdSAP) that produces an EPC. The results are in your separate **Energy Performance Report**. It has not been lodged as an EPC; it is a working baseline for this plan.

> An energy assessment works out what the house *should* cost to run if it were heated in a standard way, using the insulation, heating and glazing recorded. It does not measure whether you feel cold, where the draughts are, whether the fans work or how you use the rooms. Those things come from the thermal imaging and the inspection. The two together give a fair picture: the report sets the baseline, and the rest of this plan explains how the house actually performs.

ENERGY EFFICIENCY RATING	ENVIRONMENTAL (CO_2) RATING	ESTIMATED RUNNING COST	CO_2 SAVING AVAILABLE
69 C	**69 C**	**£874/yr**	**0.6 t/yr**
Potential 85 B	Potential 77 C	Potential £752, saving £122	Floor area 64 m²

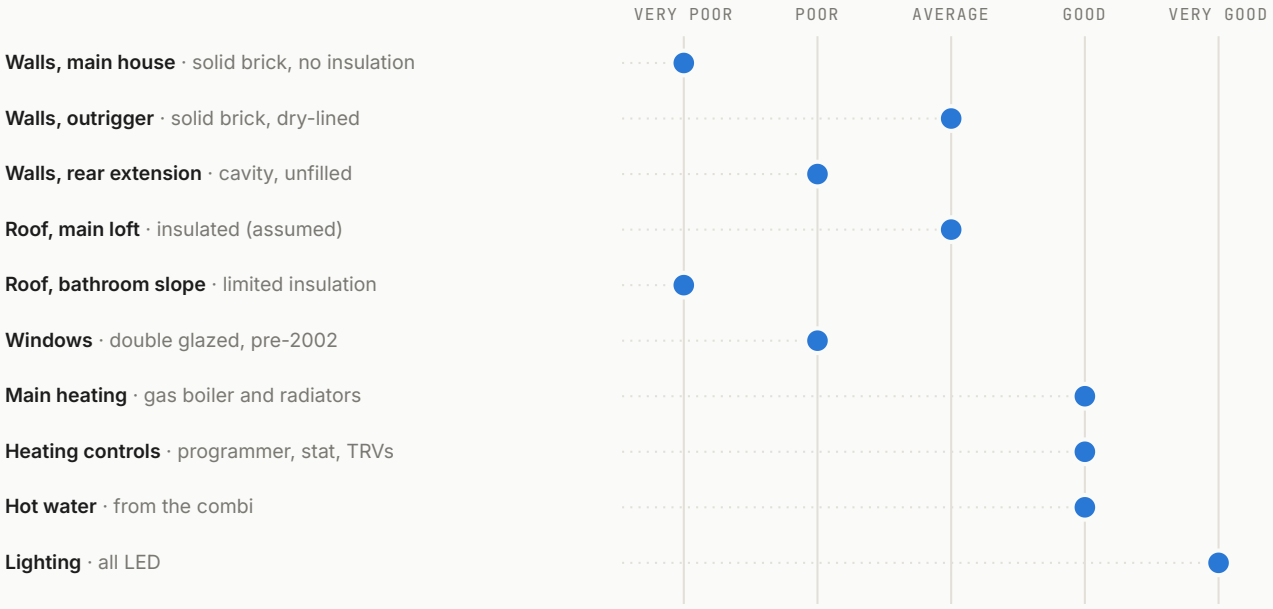
Where the house sits on the rating scale



The rating runs from 1 to 100; higher is better. The average for homes in England and Wales is band D, around 60. The 2025 EPC is shown in outline for comparison.

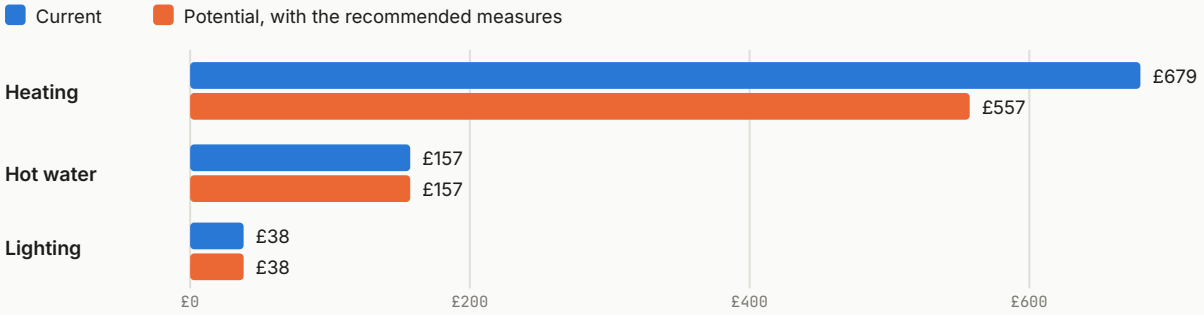
What each part of the house scores

The assessment rates each part of the house from very poor to very good for energy efficiency. This is the clearest picture of where the weak points are.



Floors are not given a rating by the method, but both the suspended timber floor and the solid floors are recorded as uninsulated. The weakest elements are the main walls and the sloping roof over the bathroom (rated on the method's default, as its insulation cannot be seen), followed by the windows and the rear extension walls. The heating, controls, hot water and lighting are already good.

Estimated running costs



Heating is three-quarters of the estimated bill, so it is where the savings are. The £122 a year saving is all on heating. These figures assume a standard heating pattern and standard fuel prices; your own bills will differ with how you heat the house.

What the terms mean

Energy efficiency rating	A score from 1 to 100 based on the estimated cost of heating, hot water and lighting per square metre. 69 puts the house at the very bottom of band C, one point above D, which is above average for an Edwardian terrace.
Environmental (CO_2) rating	The same scale, based on carbon emissions instead of cost. For this house it is also 69, in band C.
Potential rating	The score if all the recommended measures were carried out: 85 (B) for cost, 77 (C) for carbon, a saving of 0.6 tonnes of CO_2 a year.
Element ratings	How each part of the house performs compared with the best available today. "Assumed" means the method used a default for the age of the house because the insulation could not be seen.
Estimated costs	What the house would cost to run under a standard heating pattern (about 21°C in the living room and 18°C elsewhere, mornings and evenings on weekdays and most of the day at weekends). They are for comparison, not a forecast of your bills.

Why it differs from the 2025 EPC

The June 2025 EPC rated the house 66 (D). Today's assessment gives 69 (C). Nothing has been improved in between; the difference comes from what was recorded. This time the floor area was measured at 64 m² (61 m² in 2025), the rear was recorded as three separate extensions with their own wall types, and all 23 fixed light fittings are now LED. Small changes like these can move a rating by a few points, which is why the score is best used as a baseline rather than a verdict.

The report's recommendations, and how this plan treats them

RECOMMENDED BY THE ASSESSMENT	MY VIEW FOR THIS HOUSE
1. Internal wall insulation	Not now. It reduces room sizes, would disturb the new floors and decoration, and carries a real condensation risk at the party walls and the ends of the floor joists in a solid-brick house. Only with a full moisture risk assessment (Stage 3 at the earliest).
2. External wall insulation	Possible later on the rear and outrigger, using a breathable system, when the render is next due for renewal (Stage 3). Less suitable on the main front, where it would change the bay and the look of the terrace; the small porch wall is the exception (section 10).
3. Solar PV	Yes. The front slope faces south-south-east and the house has smart meters and can export. Best done after the roof repairs (Stage 2).

The method only recommends measures that change the calculated cost. It does not recommend draught-proofing, ventilation, repairs or the heating set-up, because it cannot see those problems. In this house they are the measures that will make the biggest difference to comfort, which is why the plan starts with them and not with the walls.

10 Improvement options

Every measure I considered for the house, with my view on whether it suits it. Ratings are relative to each other: ●●● high, ●●○ medium, ●○○ low. Costs are indicative 2026 ranges.

MEASURE	SUITABLE?	COMFORT	ENERGY	COST	DISRUPTION	NOTES AND RISKS
Bathroom and kitchen extract (dMEV), door undercuts	Yes, first	●●○	○○○	£450–£1,100	Low	Essential before draught-proofing. Protects the house from condensation and mould.
Roof repairs (flashing, then maintenance)	Yes, first	●○○	○○○	£800–£2,500	Low	Keeps the fabric dry. No insulation should go into a roof or wall that is getting wet.
Draught-proofing (floor edges, bays, doors, hatch)	Yes	●●●	●●○	£150–£1,200	Low	The best value measure for this house. Keep air bricks and vents clear.
Heating set-up (flow temperature, balancing, lagging)	Yes	●○○	●●○	£0–£150	None	Easy saving. Pairs with radiator upsizing.
Loft floor insulation to 270–300 mm with raised boarding	Yes, if loft stays as storage	●●○	●●○	£800–£1,800	Medium	Raised loft legs keep the storage. Keep a 50 mm air gap at the eaves. Boiler and pipes then sit in a cold loft and must be lagged.
Radiator upgrade (taller or wider, same pipework)	Optional, later	●○○	○○○	£350–£700 each	Low	The existing double convectors are reasonably sized. Only if rooms still struggle after the draughts and insulation; size from a room heat loss calculation at a 50–55°C flow temperature.
Suspended floor insulation (through-lounge)	Yes, with care	●●●	●●○	£2,000–£6,000	Medium–High	Best done from below if the void is deep enough (e.g. robotic spray or mineral wool on netting), to avoid lifting the new oak floor. Void must stay ventilated.
Insulation under the bay window (inside)	Yes, quick win	●●●	●○○	£150–£400 DIY £400–£1,000 fitted	Low	Rigid foam board (such as Celotex, up to 100 mm) cut to fit on timber battens with a small ventilated gap to the wall, plasterboard over, every joint and edge sealed so no warm air reaches the cold wall. Keep the air brick under the bay open.
Porch roof insulation	Yes, quick win	●●○	●○○	Low cost, DIY	Low	Lift the hatch and lay around 300 mm of loft roll, keeping the eaves clear for ventilation.

Porch wall insulation (outside)	Yes	●●○	●○○	£800–£2,000	Medium	Thin single-skin wall, small area. External insulation is best here, as 100 mm inside would take too much of the porch. Detail it carefully to keep rain out, and check planning if it changes the look of the front.
Replace windows and doors	Yes, medium to long term	●●●	●●○	£8,000–£14,000 whole house	Medium	Pre-2002 units with ageing frames. Specify modern A-rated units (U-value 1.4 or better) with trickle vents, and fit the patio doors and front door at the same time. TrustMark and FENSA or CERTASS installers. Can be phased, worst first.
Cavity fill (single-storey rear)	Check	●○○	●○○	£300–£600	Low	The 1967–75 rear wall is recorded as an unfilled cavity. A small area; check the cavity with a borescope before deciding, and use a system suited to its exposure.
Outrigger roof insulation (bathroom ceiling)	At re-roofing	●●○	●○○	£1,500–£3,500 extra	High	It is impossible to know whether this slope has been insulated without opening it up. Check, and insulate from above if needed, when the slates are next replaced, rather than disturbing the new bathroom. Replace the rooflight at the same time.
Rear extension roof insulation	At re-roofing	●○○	●○○	£1,500–£3,000 extra	High	The roof is newer and looked better on the thermal images, so it may already have some insulation. Warm-roof upgrade when the covering reaches the end of its life.
Solar PV on the front roof slope	Yes	○○○	●●●	£5,500–£8,000	Low	Recommended by the energy assessment. The front slope faces SSE, which is close to ideal, and smart meters are fitted. Check shading from the chimney and No 94's dormer; around 3–4 kWp likely. Do after the roof repairs.
Air source heat pump	Later	●○○	●●●	£10,000–£14,000 before grant	Medium	Feasible once the fabric is improved and radiators upsized. £7,500 Boiler Upgrade Scheme grant currently available. Needs space for the outdoor unit at the rear.

External wall insulation (rear and outrigger)	Possible, later	●●○	●●○	£8,000–£15,000	High	Must be a breathable system (mineral wool or wood fibre, lime or silicate render) with careful detailing at the party walls, eaves and windows. A BS 5250 moisture risk assessment is needed.
Internal wall insulation (front)	Not recommended now	●●○	●○○	£6,000–£10,000	High	Loses room space, disturbs the new floors and decoration, and carries condensation risk at the party walls and floor joists. Revisit only with a full moisture risk assessment.
Loft conversion (habitable room)	Your choice	●●●	●○○	£35,000+	High	Only route to a properly warm loft room. Requires Building Regulations approval, roof insulation to current standards, fire escape and a fixed staircase.

11 Your medium-term improvement plan

This is the order I would do things in. Each stage makes the next one safer or more effective: first keep the house dry, then stop the draughts, then add insulation, and only then change how the house is heated. Later stages are tied to natural trigger points, such as re-roofing, so you are not paying twice for scaffolding or making good.

Stage 0

NOW · BEFORE WINTER
REPAIRS AND
VENTILATION

MEASURE	COST
Replace the bathroom fan with a dMEV; add a dMEV or ducted hood to the kitchen; fit a cowl to the outrigger wall grille	£450–£1,000
Trim internal doors to a 10 mm undercut; add a trickle vent to bedroom 1 (and ideally the front bay)	DIY or £20–£40 a door
Re-fix the loose flashing at the front; check the other side	£250–£800
Keep the air bricks under the bay and at the back clear of decking and planters	DIY
Lag loft pipework and the boiler condensate pipe	£20–£60 DIY

Stage 1

THIS WINTER
DRAUGHTS AND QUICK
WINS

MEASURE	COST
Insulate under the bay window: rigid foam on battens, plasterboard, all joints sealed	£150–£1,000
Insulate the porch roof: around 300 mm through the hatch	Low cost, DIY
Draught-proof skirtings, window frames, front door, letterbox, inner porch door and loft hatch; clean, re-brush and adjust the patio doors	£150–£1,200
Turn the boiler flow temperature down to 55–60°C; balance the radiators at the next service	£0–£150
Decide on the loft. If storage: top up the floor insulation to 270–300 mm on raised boarding and switch off the loft radiator	£800–£1,800
Kitchen reveal: strip, dry out and redecorate with breathable paint	DIY

Stage 2

1–3 YEARS
WINDOWS, PORCH AND FLOORS

MEASURE	COST
Replace windows and doors, worst first if phased	£8,000–£14,000
External insulation to the thin porch wall	£800–£2,000
Roof maintenance: chipped tiles, parapet capping, chimney flaunching, lichen treatment	£550–£1,700
Insulate the suspended floor in the through-lounge, from below if the void allows, or when the floor is next replaced	£2,000–£6,000
Optional: upgrade downstairs radiators if rooms still struggle	£350–£700 each
Check the rear extension cavity and fill it if suitable	£300–£600
Repair the render crack at the bay cornice with a compatible, breathable repair	£150–£400
Solar PV on the front roof slope, after the roof repairs	£5,500–£8,000

Stage 3

3–10 YEARS
AT TRIGGER POINTS

MEASURE	TRIGGER
Insulate the outrigger roof from above and replace the rooflight	Re-roofing the outrigger
Warm-roof upgrade to the single-storey rear	Re-roofing the rear
External wall insulation to the rear and outrigger	Re-rendering the rear
Air source heat pump with Boiler Upgrade Scheme grant	Boiler replacement
Loft conversion, if wanted, insulated to current standards	Your plans

> Each stage should be checked against the one before it. The small Stage 1 insulation jobs (under the bay and the porch roof) are fine once the new fans are in. Leave the larger ones (loft, floor, walls) until the house has been through a winter with the fans running, so that any remaining damp shows itself first.

12 What to avoid on a solid-brick house

Solid walls stay dry by letting moisture out as well as in. Most problems in houses like yours come from well-meant work that traps moisture. Please avoid:

- **Spray foam on the underside of the roof.** It hides the condition of the timbers, can trap moisture against them, and often causes problems with mortgages and sales.
- **Cement renders, plastic masonry paints and gypsum-based plasters** on the solid walls. Use lime, silicate or other breathable materials for repairs.
- **Blocking air bricks** or covering them with decking, soil or planters.
- **Jet-washing the roof tiles.** It strips the surface and shortens their life. Use a biocide treatment and gentle brushing.
- **Upgrading one thing in isolation** without checking its effect on the rest: for example, new airtight windows without trickle vents, or insulation that stops short and leaves a cold strip where mould will form.

13

Funding and next steps

Funding

- **Boiler Upgrade Scheme:** currently £7,500 towards an air source heat pump, through an MCS-certified installer.
- **0% VAT** currently applies to the installation of energy-saving materials such as insulation, solar PV and heat pumps.
- **Local and government schemes** (such as Warm Homes local grants and ECO) depend on household income and change regularly. I am happy to check eligibility before you commit to any larger measure.

Next steps

- Read through the plan and the Energy Performance Report, then give me a call if you would like to talk any of it through. That is included in the fee.
- Get quotes for Stage 0 now. I can review any quotes or specifications against this plan.
- Larger measures (floor insulation, wall insulation, heat pump) should be designed and installed by TrustMark-registered contractors, with a retrofit coordinator overseeing the work where it is funded under PAS 2035.

Assessor

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Limitations

This plan is advisory. It sets out measures that appear suitable for the house, the order to do them in and the risks to manage; it is not a design, a specification or an instruction to carry out work. It is not an Energy Performance Certificate and has not been lodged on any register. The inspection was visual and non-invasive: I did not lift floor coverings or boards, open up walls or test services. The roof was viewed by drone and the loft from its boarded floor; areas hidden by stored items, boarding or decoration could not be inspected. Thermal images record surface temperatures on the morning of the visit and are affected by weather, sun and how the house was heated; they indicate where heat is lost and are interpreted alongside the visual inspection. Costs are indicative 2026 ranges for budgeting only. The plan is prepared for the client under the Terms and Conditions of Engagement dated 8 October 2026 and should not be relied on by anyone else.